Project Title	**🎯 Job Acceptance Prediction System**
Skills take away From This Project	• **Python for Data Analytics (Pandas, NumPy)** • **Data Cleaning & Preprocessing** • **Handling Missing & Noisy Real-World Data** • **Exploratory Data Analysis (EDA)** • **Feature Engineering for Machine Learning** • **Supervised Machine Learning (Classification)** • **Model Evaluation & Performance Metrics**
Domain	• **Human Resource Analytics (HR Analytics) using Machine Learning** • **HR Analytics & Predictive Modeling (Machine Learning)** • **Recruitment & Placement Analytics – Machine Learning**

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Problem Statement:

Recruitment and placement teams deal with large volumes of candidate data related to academic performance, skills, experience, interview outcomes, and job market conditions.
However, not all candidates who are eligible or receive offers end up accepting them.

The goal of this project is to **analyze candidate placement data and build a Job Acceptance Prediction System** to:

- Predict whether a candidate will **accept or reject a job offer**
- Identify key factors influencing job acceptance decisions
- Handle real-world data challenges such as **missing values and noisy data**
- Provide actionable insights to improve recruitment and placement strategies

Business Use Cases:

- Help recruitment teams predict offer acceptance probability
- Reduce offer dropouts and hiring cycle time
- Identify factors impacting placement success
- Optimize interview and evaluation strategies
- Support data-driven decision-making for HR managers

🔍 Approach

🔹 Step 1: Data Collection

- **Dataset consists of 50,000 candidate records**

- **Closely follows real-world recruitment patterns**
- **Includes intentional data issues such as:**
    - **Missing values**
    - **Duplicate-like records**
    - **Inconsistent categorical values**

🔹 Step 2: Data Understanding

Dataset contains features covering:

- **Academic performance**
- **Skills and certifications**
- **Work experience**
- **Interview performance**
- **Job market conditions**
- **Placement outcome (target)**

Initial checks performed:

- **Dataset shape and size**
- **Data types validation**
- **Sample record inspection**

- **Null value distribution analysis**

🔹 Step 3: Data Cleaning & Preprocessing

Key preprocessing steps include:

- **Handling missing values using mean, median, or mode**
- **Encoding categorical variables (Label / One-Hot Encoding)**
- **Feature scaling for numerical columns**
- **Correcting inconsistent categorical labels**
- **Ensuring logical consistency across features**

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🔹 Step 4: Exploratory Data Analysis (EDA)

EDA is performed to uncover patterns and trends such as:

- **Interview score vs job acceptance**
- **Skills match percentage impact on placement**
- **Company tier vs acceptance rate**
- **Experience vs placement probability**
- **Competition level vs job acceptance**
- **Correlation analysis among numeric features**

🔹 Step 5: Feature Engineering

Derived analytical features include:

- **Experience category (Fresher / Junior / Senior)**
- **Academic performance bands**
- **Skills match level (Low / Medium / High)**
- **Interview performance category**
- **Placement probability score**

🔹 Step 6: Data Storage (Optional – SQL)

- **Cleaned dataset stored in MySQL**
- **Tables created with appropriate data types**
- **Data inserted using SQLAlchemy**
- **Enables scalable querying and reporting**

🔹 Step 7: Machine Learning Modeling

Business-focused ML modeling includes:

- **Defining status as the target variable**
    - **Placed → Job Accepted (1)**
    - **Not Placed → Job Rejected (0)**

8) Analyst Tasks (EDA & ML Analytics)

Candidate Performance Analysis

- **Academic scores vs placement outcome**
- **Skills match vs interview performance**
- **Certification impact on job acceptance**

Placement & Acceptance Analysis

- **Acceptance rate by company tier**
- **Experience vs placement success**

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Interview & Evaluation Analysis

- **Interview score vs placement probability**
- **Employability test score analysis**

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9) Key KPIs in Streamlit dashboards.

- **Total Candidates**
- **Placement Rate (%)**
- **Job Acceptance Rate (%)**

- **Average Interview Score**
- **Average Skills Match %**
- **Offer Dropout Rate**
- **High-Risk Candidate Percentage**

✅ Results

- A cleaned and normalized **job placement dataset** with missing values handled and inconsistencies corrected.
- A structured analytical dataset containing academic, skill-based, experiential, and interview-related features.
- Machine learning models capable of predicting **job acceptance with strong accuracy and explainability**.
- Actionable insights highlighting the **key drivers of placement success**.
- Comprehensive documentation covering preprocessing, modeling, insights, and recommendations.

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📊 Project Evaluation Metrics

Data Cleaning & EDA Accuracy (25%)

- Quality of missing value handling
- Correct preprocessing and feature engineering
- Logical consistency checks

Machine Learning (45%)

- Model correctness and evaluation
- Feature importance interpretation
- Business relevance of insights

Visualization & Dashboard (10%)

- Clarity and interactivity of Streamlit dashboard
- Effective representation

Documentation & Communication (10%)

- Clear explanation of dataset, approach, and insights
- Professional presentation and storytelling

Project Organization & Reproducibility (10%)

- Clean folder structure
- Readable and reusable code
- Reproducible results using provided datasets

Data Set:
https://drive.google.com/file/d/1h7IY0sNyW7tDoI03nO4lohiYk04o0Hqd/view?usp=sharing

Data Set Explanations :

Dataset_Explanations.

Project Deliverables:

📦 Project Deliverables

- **Source Code:** Python scripts for EDA, preprocessing, and ML modeling
- **SQL Database:** Cleaned dataset stored in MySQL
- **Streamlit :** Interactive dashboards
- **Documentation:** Detailed project report

References:

Project Live Evaluation Metrics	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) Guide
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Streamlit - Tamil - Recordings	Streamlit - Tamil
Streamlit - English - Recordings	Special session for STREAMLIT
HOW TO ESTABLISH SQL CONNECTION	PYTHONSQLCODE_TAMIL.ipynb

Project Orientation (Tamil)	Project Orientation Session : Job A…
Project Orientation (English)	

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:30PM to 4:30PM)

Booking link : **Project Doubt Session [DS/AIML]**

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:
For DS and AIML
Monday-Saturday (05:30PM to 07:00PM)

- ***Booking link :*** **https://forms.gle/1m2Gsro41fLtZurRA**

Evaluation Metrics *:* **Project Live Evaluation Metrics**

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